

Designing for Human, Digital, and Physical Interplay in Mixed Reality

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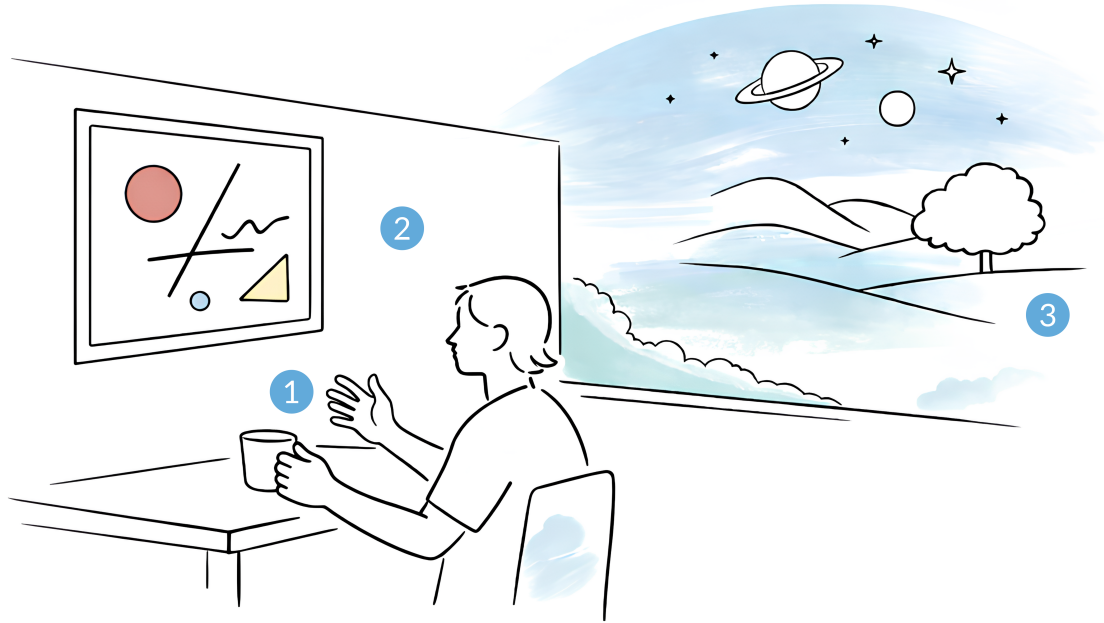


Figure 1: A conceptual illustration of the research vision. The interaction continuum expands from immediate tabletop interactions with objects and gestures [5] ❶, to surrounding boundaries and surfaces [6] ❷, and ultimately to broader environments and spaces beyond the here and now ❸.

Abstract

This extended abstract outlines a research trajectory for mixed reality that expands human awareness outward—from intimate interactions with everyday objects to an embodied connection with the cosmos. It advocates for spatial computing paradigms that profoundly deepen, rather than displace, our vital grounding in the physical world.

CCS Concepts

• **Human-centered computing** → **Interaction design**; **Mixed / augmented reality**.

Keywords

research through design, spatial interaction design

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1 Vision

「仰觀宇宙之大，俯察品類之盛，所以遊目騁懷，足以極視聽之娛。」¹

— 王羲之(Wang Xizhi), 353 AD

What does it mean to be truly spatial? As mixed reality (MR) becomes ubiquitous, we are witnessing an unprecedented interplay

¹English translation: “Looking up, we marvel at the vastness of the universe; looking down, we examine the multitude of physical things. Thus we let our eyes roam and our minds wander, fully enjoying the pleasures of our senses.”



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between digital elements, the physical environment, and human participation. I view this interplay as the convergence of two distinct trajectories—MR and tangible user interfaces (TUIs)—destined to intersect. Ultimately, I believe humanity will engineer Digital Matter: entities as tangible as physical objects [3], yet as mutable and transient as digital pixels [2].

While the technology to fully realize Digital Matter does not yet exist, designing for this future poses a question: When we possess the capability to construct realities, does it mean abandoning our physical one? My research journey posits the opposite. As our digital capabilities expand, it becomes increasingly vital that humans remain deeply grounded in the physical reality. This grounding is not merely about simulating the real world; it is about utilizing the full spectrum of human senses to establish profound connections with our environment.

2 Framework

I see the future of human-digital-physical interplay as a journey of perception. It is a process of expanding human awareness outward from the immediate self, stretching across a spectrum from the “here and now” to the “not here, not now” [1, 4]. My research samples critical points along this expanding continuum.

The Immediate

The journey begins within our immediate surroundings. Everyday objects are more than functional tools; they are tangible reflections of human preferences, habits, and lifestyles. My initial research investigates how to weave these physical artifacts into spatial interactions [5].

By combining the natural familiarity of everyday objects as physical inputs with the fluidity of mid-air gestures, I identified five interaction types that form a design space for expressive uni- and bimanual interaction. Through a user study, I demonstrated that they support basic 3D manipulation with performance comparable to the status quo. The case studies further revealed that users found them intuitive and were interested in integrating them into their daily lives.

The Enclosed

Expanding our awareness outward, we encounter the surfaces and boundaries of our enclosed spaces—rooms, offices, and architecture designed to protect and contain us. Moving beyond the immediate grasp of the desktop, I explore how we might interact with the macro-geometry of our environment [6].

If future Digital Matter can defy physical laws, a fascinating property emerges: it can become unbounded, capable of penetrating and traversing physical walls. By employing a Research through Design [8] inquiry featuring design space iterations, demonstrations, and expert feedback sessions, I expanded the scope of object-boundary interactions and highlighted how they can enrich and reframe the spatial experience.

The Expansive

Moving beyond static, enclosed spaces, the spectrum of awareness extends into the broader, dynamic world. Here, my ongoing research questions the dominance of visual and haptic feedback. How can

other senses be materialized? Consider auditory interplay: What if sound could be transformed into tangible matter? What does it mean for a user to take a physical knife to “slice” a sound, or to “smear” it?

Furthermore, this broader environment introduces the complexity of relative motion. When a user is inside a moving vehicle—a spatial frame shifting within a larger spatial frame—how does this relative movement unlock novel paradigms for interplay?

The Cosmic

Ultimately, to truly stretch the boundaries of spatial interplay, we may look up. In an era of urban light pollution, our visual and emotional connection to the universe has been severed. Reconnecting with the cosmos evokes a profound sense of humility, reminding us of our fragile, microscopic existence within the grander scale of reality.

My future exploration asks how we can design interactions with the stars and the outer universe to foster this vital humility. By adopting a more-than-human [7] perspective—such as using MR to temporarily embody a planet—we could gain a sobering, expansive awareness of our existence and what we mean to one another as a species.

3 In Closing

To design for the future of mixed reality is to reflect on this entire spectrum of awareness—from the physical coffee cup on our desk to the distant orbit of a planet. The physical world provides a vast, nearly infinite wellspring of inspiration. As we design and build new spatial realities, we should connect with this foundational reality rather than overwrite or erase it. I invite the community to join in designing a future where technology deepens such connections, approaching our intertwined existence with the universe with the awe and humility it demands.

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